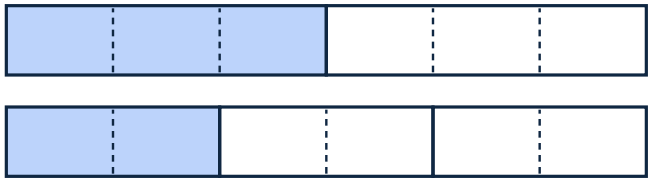
# Unlike Fractions

Different bottoms? Cut the pieces until they match, then add or take away.

### 1/2 + 1/3: halves and thirds don't match



Cut each half into 3: sixths.
Cut each third into 2: sixths.
Now the pieces match.

$$\frac{1}{2} + \frac{1}{3} = \frac{3}{6} + \frac{2}{6} = \frac{5}{6}$$

### Find the same size pieces

Count by the bigger bottom until the other bottom fits: 4, 8, 12. 3 fits in 12.

$$\frac{2}{3} - \frac{1}{4} = \frac{8}{12} - \frac{3}{12} = \frac{5}{12}$$

### Mixed numbers: wholes, then pieces

$$1\ \frac{1}{2} + 2\ \frac{1}{3} = 3\ \frac{5}{6}$$

1 + 2 = 3 wholes.
1/2 + 1/3 = 5/6.

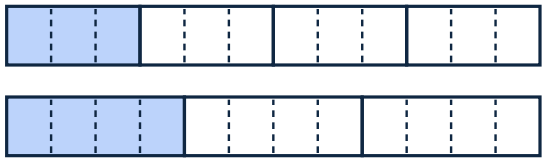
**Your turn!**

$$\frac{1}{4} + \frac{1}{6} = \frac{\square}{12} + \frac{\square}{12} = \frac{\square}{12}$$

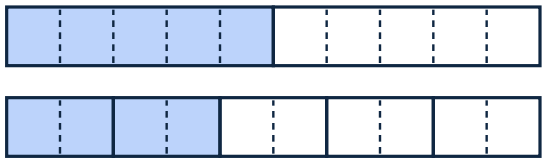
# Unlike Fractions: Practice

Same size pieces first. Then add or take away the tops.

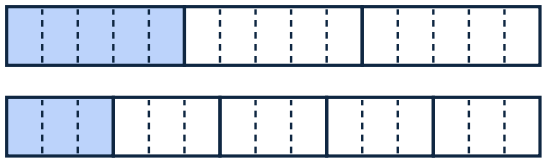
**Part 1** Cut the pieces The dashed lines make the same size pieces.

1) 

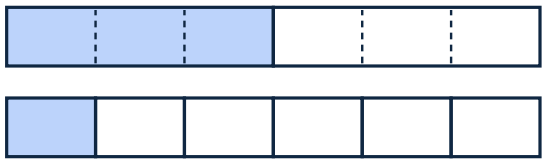
$$\frac{1}{4} + \frac{1}{3} = \frac{\square}{12} + \frac{\square}{12} = \frac{\square}{12}$$

2) 

$$\frac{1}{2} + \frac{2}{5} = \frac{\square}{10} + \frac{\square}{10} = \frac{\square}{10}$$

3) 

$$\frac{1}{3} + \frac{1}{5} = \frac{\square}{15} + \frac{\square}{15} = \frac{\square}{15}$$

4) 

$$\frac{1}{2} + \frac{1}{6} = \frac{\square}{6} + \frac{\square}{6} = \frac{\square}{6}$$

**Part 2** Same bottoms The new bottom is given.

5) $\frac{1}{3} + \frac{3}{4} = \frac{\square}{12} + \frac{\square}{12} = \frac{\square}{12}$

6) $\frac{1}{2} + \frac{3}{8} = \frac{\square}{8} + \frac{\square}{8} = \frac{\square}{8}$

7) $\frac{5}{6} - \frac{1}{3} = \frac{\square}{6} - \frac{\square}{6} = \frac{\square}{6}$

## Unlike Fractions: Practice (continued)

### Part 2 Same bottoms (continued)

8) $\frac{2}{5} + \frac{1}{4} = \frac{\square}{20} + \frac{\square}{20} = \frac{\square}{20}$

9) $\frac{7}{10} - \frac{1}{2} = \frac{\square}{10} - \frac{\square}{10} = \frac{\square}{10}$

10) $\frac{1}{6} + \frac{4}{9} = \frac{\square}{18} + \frac{\square}{18} = \frac{\square}{18}$

11) $\frac{3}{4} + \frac{1}{10} = \frac{\square}{20} + \frac{\square}{20} = \frac{\square}{20}$

12) $\frac{5}{8} - \frac{1}{4} = \frac{\square}{8} - \frac{\square}{8} = \frac{\square}{8}$

### Part 3 Add No help this time. Find the same size pieces.

13) $\frac{1}{2} + \frac{1}{5} = \frac{\square}{\square}$

14) $\frac{2}{3} + \frac{1}{4} = \frac{\square}{\square}$

15) $\frac{1}{4} + \frac{3}{5} = \frac{\square}{\square}$

16) $\frac{3}{8} + \frac{1}{3} = \frac{\square}{\square}$

17) $\frac{1}{6} + \frac{2}{3} = \frac{\square}{\square}$

18) $\frac{2}{5} + \frac{1}{2} = \frac{\square}{\square}$

## Unlike Fractions: Practice (continued)

### Part 3 Add (continued)

19) $\frac{1}{2} + \frac{3}{7} = \frac{\square}{\square}$

20) $\frac{3}{4} + \frac{2}{3} = \frac{\square}{\square}$

21) $\frac{1}{3} + \frac{2}{9} = \frac{\square}{\square}$

22) $\frac{5}{6} + \frac{1}{8} = \frac{\square}{\square}$

23) $\frac{3}{10} + \frac{1}{4} = \frac{\square}{\square}$

24) $\frac{2}{7} + \frac{1}{3} = \frac{\square}{\square}$

25) $\frac{1}{5} + \frac{3}{8} = \frac{\square}{\square}$

26) $\frac{4}{9} + \frac{1}{6} = \frac{\square}{\square}$

27) $\frac{1}{8} + \frac{5}{12} = \frac{\square}{\square}$

28) $\frac{3}{5} + \frac{1}{6} = \frac{\square}{\square}$

## Unlike Fractions: Practice (continued)

**Part 4** Subtract No help this time. Find the same size pieces.

29) $\frac{3}{4} - \frac{1}{3} = \frac{\square}{\square}$

30) $\frac{5}{6} - \frac{1}{4} = \frac{\square}{\square}$

31) $\frac{2}{3} - \frac{1}{5} = \frac{\square}{\square}$

32) $\frac{7}{8} - \frac{1}{6} = \frac{\square}{\square}$

33) $\frac{4}{5} - \frac{1}{2} = \frac{\square}{\square}$

34) $\frac{5}{8} - \frac{1}{3} = \frac{\square}{\square}$

35) $\frac{9}{10} - \frac{1}{4} = \frac{\square}{\square}$

36) $\frac{3}{4} - \frac{2}{7} = \frac{\square}{\square}$

37) $\frac{5}{6} - \frac{2}{9} = \frac{\square}{\square}$

38) $\frac{2}{3} - \frac{1}{8} = \frac{\square}{\square}$

39) $\frac{7}{10} - \frac{1}{3} = \frac{\square}{\square}$

40) $\frac{4}{9} - \frac{1}{6} = \frac{\square}{\square}$

## Unlike Fractions: Practice (continued)

### Part 4 Subtract (continued)

41) $\frac{7}{12} - \frac{1}{8} = \frac{\square}{\square}$

42) $\frac{3}{5} - \frac{1}{4} = \frac{\square}{\square}$

43) $\frac{6}{7} - \frac{1}{2} = \frac{\square}{\square}$

44) $\frac{5}{9} - \frac{1}{6} = \frac{\square}{\square}$

### Part 5 Mixed numbers No help this time. Wholes, then pieces.

45) $1\frac{1}{2} + 2\frac{1}{4} = \square\ \frac{\square}{\square}$

46) $2\frac{1}{3} + 1\frac{1}{2} = \square\ \frac{\square}{\square}$

47) $3\frac{1}{4} + 1\frac{2}{3} = \square\ \frac{\square}{\square}$

48) $1\frac{2}{5} + 2\frac{1}{2} = \square\ \frac{\square}{\square}$

49) $4\frac{1}{6} + 1\frac{1}{4} = \square\ \frac{\square}{\square}$

50) $3\frac{2}{3} - 1\frac{1}{2} = \square\ \frac{\square}{\square}$

51) $5\frac{3}{4} - 2\frac{1}{3} = \square\ \frac{\square}{\square}$

52) $4\frac{5}{6} - 1\frac{1}{4} = \square\ \frac{\square}{\square}$

## Unlike Fractions: Practice (continued)

### Part 5 Mixed numbers (continued)

53) $6\frac{1}{2} - 3\frac{1}{5} = \square\frac{\square}{\square}$

### Part 6 Add

54) $\frac{3}{5} + \frac{1}{2} = \frac{\square}{\square}$

55) $\frac{2}{4} + \frac{1}{3} = \frac{\square}{\square}$

56) $\frac{5}{6} + \frac{5}{8} = \frac{\square}{\square}$

57) $\frac{2}{4} + \frac{4}{5} = \frac{\square}{\square}$

58) $\frac{1}{6} + \frac{1}{4} = \frac{\square}{\square}$

59) $\frac{6}{8} + \frac{2}{5} = \frac{\square}{\square}$

60) $\frac{1}{3} + \frac{1}{5} = \frac{\square}{\square}$

61) $\frac{3}{4} + \frac{2}{5} = \frac{\square}{\square}$

62) $\frac{1}{8} + \frac{2}{3} = \frac{\square}{\square}$

63) $\frac{2}{8} + \frac{2}{5} = \frac{\square}{\square}$

## Unlike Fractions: Practice (continued)

### Part 6 Add (continued)

64) $\frac{1}{5} + \frac{1}{8} = \frac{\square}{\square}$

65) $\frac{2}{3} + \frac{3}{4} = \frac{\square}{\square}$

66) $\frac{1}{4} + \frac{2}{3} = \frac{\square}{\square}$

67) $\frac{3}{4} + \frac{1}{3} = \frac{\square}{\square}$

68) $\frac{1}{2} + \frac{1}{3} = \frac{\square}{\square}$

69) $\frac{3}{6} + \frac{3}{4} = \frac{\square}{\square}$

70) $\frac{2}{3} + \frac{1}{2} = \frac{\square}{\square}$

71) $\frac{3}{6} + \frac{2}{5} = \frac{\square}{\square}$

### Part 7 Subtract

72) $\frac{3}{4} - \frac{4}{6} = \frac{\square}{\square}$

73) $\frac{4}{5} - \frac{2}{4} = \frac{\square}{\square}$

74) $\frac{3}{5} - \frac{1}{3} = \frac{\square}{\square}$

75) $\frac{1}{3} - \frac{1}{4} = \frac{\square}{\square}$

## Unlike Fractions: Practice (continued)

### Part 7 Subtract (continued)

76) $\frac{2}{3} - \frac{3}{5} = \frac{\square}{\square}$

77) $\frac{4}{5} - \frac{1}{8} = \frac{\square}{\square}$

78) $\frac{2}{3} - \frac{1}{4} = \frac{\square}{\square}$

79) $\frac{3}{4} - \frac{2}{5} = \frac{\square}{\square}$

80) $\frac{2}{4} - \frac{1}{6} = \frac{\square}{\square}$

81) $\frac{1}{4} - \frac{1}{5} = \frac{\square}{\square}$

82) $\frac{3}{6} - \frac{1}{5} = \frac{\square}{\square}$

83) $\frac{1}{3} - \frac{1}{5} = \frac{\square}{\square}$

84) $\frac{2}{5} - \frac{1}{6} = \frac{\square}{\square}$

85) $\frac{1}{2} - \frac{1}{3} = \frac{\square}{\square}$

86) $\frac{5}{6} - \frac{2}{4} = \frac{\square}{\square}$

87) $\frac{3}{8} - \frac{1}{6} = \frac{\square}{\square}$

## Unlike Fractions: Practice (continued)

### Part 7 Subtract (continued)

88) $\frac{1}{2} - \frac{2}{5} = \frac{\square}{\square}$

89) $\frac{6}{8} - \frac{4}{6} = \frac{\square}{\square}$

### Part 8 Remember? Problems from earlier packets.

90) $7 \times 0.6 = \square$

91) $3.6 \div 4 = \square$

92) $0.7 \times 0.3 = \square$

93) $945 \div 35 = \square$

94) $\frac{5}{6} = \frac{\square}{18}$

95) $\frac{3}{8} \square \frac{1}{2}$

### Part 9 Word problems Same size pieces first.

96) Mia walked 1/2 mile to school and 1/3 mile to the park. How far did she walk?

Number sentence _____

Answer ☐ mile

Draw or show your work

97) A jug had 3/4 L of milk. Ben drank 1/3 L. How much milk is left?

Number sentence _____

Answer ☐ L

Draw or show your work

98) Leo read 2/5 of a book on Monday and 1/4 of it on Tuesday. How much did he read?

Number sentence _____

Answer ☐ of the book

Draw or show your work

## Unlike Fractions: Practice (continued)

### Part 9 Word problems (continued)

99) A board is 2 1/2 m long. Zoe cuts off 1 1/3 m. How long is the rest?

Number sentence _____

Answer ☐ m

Draw or show your work

100) Challenge: Ava had 1 whole pizza. She ate 1/4 and Sam ate 1/3. How much is left?

Number sentence _____

Answer ☐ of the pizza

Draw or show your work

# Check-In: Unlike Fractions

Show what you know. No help this time. Score: _____ / 11

**Part 1** Same bottoms The new bottom is given.

1) $\frac{3}{8} + \frac{1}{6} = \frac{\square}{24} + \frac{\square}{24} = \frac{\square}{24}$

**Part 2** Add

2) $\frac{1}{2} + \frac{1}{7} = \frac{\square}{\square}$

3) $\frac{3}{4} + \frac{1}{5} = \frac{\square}{\square}$

4) $\frac{1}{6} + \frac{2}{5} = \frac{\square}{\square}$

5) $\frac{1}{3} + \frac{1}{4} = \frac{\square}{\square}$

**Part 3** Subtract

6) $\frac{5}{8} - \frac{1}{6} = \frac{\square}{\square}$

7) $\frac{4}{5} - \frac{1}{3} = \frac{\square}{\square}$

8) $\frac{3}{4} - \frac{1}{5} = \frac{\square}{\square}$

9) $\frac{2}{3} - \frac{1}{2} = \frac{\square}{\square}$

**Part 4** Mixed numbers

10) $4\frac{2}{3} - 2\frac{1}{4} = \square\frac{\square}{\square}$

11) $1\frac{1}{6} + 3\frac{1}{4} = \square\frac{\square}{\square}$